Microstates and Macroscopic Thermodynamics of a Binary System

Statistical Physics — Laboratory Practice 1

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Abstract

A system of N independent unbiased coins is used as an exact model of a two-state assembly. All 2^N microstates are enumerated with a bitwise generator (memory $O(N)$, one counter per macrostate); the time doubles with every added coin, which limits full enumeration to $N \approx 31$ in five minutes, while storing every state exhausted the RAM at $N = 28$ ($N = 27$ completed in 104 s). The multiplicity $\Omega(N, k)$ is computed in log-space with the log-gamma function, which removes the double-precision overflow that stops direct evaluation at $N \approx 10^3$. For the team size $N = 138$ the most probable macrostate has $P_{69} = 0.068$ and 92.7% of the probability lies within ± 10 heads of it; the ordered macrostate with 20 heads has $38.2\, k_\mathrm{B}$ less entropy, and reaching 8 heads by random tossing takes $\sim 4 \times 10^{21}$ years. For $N \sim 10^{23}$ the distribution is a Gaussian of relative width 3.2×10^{-12}, so the Second Law becomes deterministic. Code and tests: `https://github.com/100542175/LAB-1-STATISTICAL-PHYSICS`.

1 Introduction

A *microstate* is the configuration of all N coins (e.g. HHTHT); a *macrostate* is fixed by the number of heads k and contains

$$\Omega(N, k) = \binom{N}{k} = \frac{N!}{k!\,(N-k)!}, \qquad \sum_{k=0}^{N} \Omega(N, k) = 2^N \tag{1}$$

equiprobable microstates. The probability of a macrostate is $P_k = \Omega(N, k)/2^N$ and its Boltzmann entropy is $S = k_\mathrm{B} \ln \Omega$. We study the computational limits of enumerating microstates and of evaluating $\Omega(N, k)$ (Part 1), the probabilities, entropy changes and waiting times for a team-specific $N = 100 + M$ (Part 2), and the extrapolation to $N \sim 10^{23}$ (Sections 4.3 and 4.4).

2 Methodology

The code is written in Python in a Conda environment (NumPy, SciPy, Matplotlib, pytest) and organised in modules (`combinatorics`, `microstates`, `thermodynamics`, `plots`), with unit tests and a `main.py` that reproduces every figure and number. Microstates are integers $s \in [0, 2^N)$ (bit $i = 1$: coin i is heads), generated with `yield` up to `1 << N`; the number of heads is the population count of s. Large multiplicities are computed in log-space,

$$\ln \Omega(N, k) = \ln \Gamma(N + 1) - \ln \Gamma(k + 1) - \ln \Gamma(N - k + 1), \tag{2}$$

and validated with `pytest` against exact integers (`math.comb`) for $N \leq 199$ and all k. Probabilities and entropies are exact calculations, so their only uncertainty is floating-point rounding, quantified by this comparison; timings are mean ± standard deviation of repeated runs. All quantities are dimensionless unless a unit is given.

3 Part 1: computational analysis of microstates

3.1 Enumeration for $N = 7$

With $2^7 = 128$ microstates and $N + 1 = 8$ macrostates, Table 1 lists the multiplicities obtained by enumerating all states with the program; every microstate is listed in Appendix A.

heads k	tails $7 - k$	$\Omega(7, k)$	example microstate
7	0	1	HHHHHHH
6	1	7	HHHHHHT
5	2	21	HHHHHTT
4	3	35	HHHHTTT
3	4	35	HHHTTTT
2	5	21	HHTTTTT
1	6	7	HTTTTTT
0	7	1	TTTTTTT

Table 1: Macrostates of $N = 7$ coins (Ω dimensionless, exact integers, no uncertainty; sum $= 128 = 2^7$).

3.2 Largest N with full enumeration

The number of microstates doubles with each added coin, $2^{N+1}/2^N = 2$, so the cost is exponential in both time and memory. **Storing every state** as a string needs $\propto N\,2^N$ bits; in CPython each string object costs $\sim N+40$ bytes plus an 8-byte list pointer, i.e. ≈ 10 GB at $N = 27$ and ≈ 21 GB at $N = 28$ (estimate). Measured with this approach, $N = 27$ ($2^{27} = 134\,217\,728$ states) completed in 104.3 s, while $N = 28$ stopped because the RAM was exhausted. The limit is therefore *memory*, not time.

Generator. Visiting the states with a generator and keeping only one counter per macrostate needs $O(N)$ memory (peak ≈ 0.7 kB versus 0.65 MB for storing all states at $N = 14$), so the time becomes the limit. The measured time grows as 2^N (Fig. 1: ≈ 0.5 s at $N = 22$), and extrapolating this scaling to the five-minute budget gives $N_{\max} \approx 31$ (extrapolated, not measured up to that value).

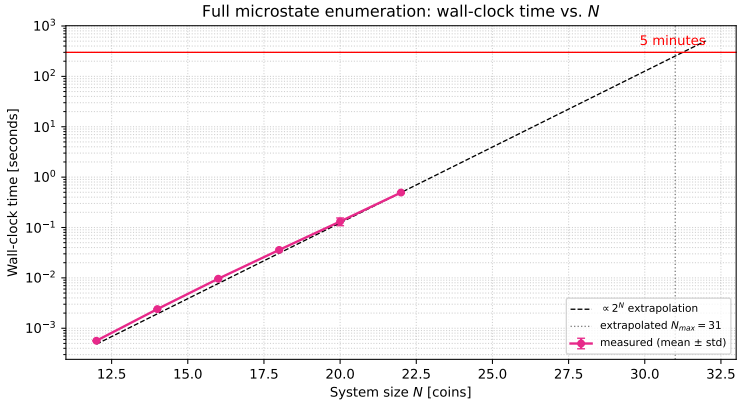


Figure 1: Wall-clock time (s) of the full enumeration versus N (mean ± std of repeated runs, log scale). Dashed: $\propto 2^N$ extrapolation; red: five minutes; dotted: extrapolated $N_{\max} = 31$.

3.3 Macrostate distributions for $N = 20, 50, 80, 100$

Figure 2 shows $P_k = \Omega(N,k)/2^N$ (left) and $\Omega(N,k)$ on a logarithmic scale (right) for the four sizes. All distributions are symmetric about $k = N/2$ (all N are even) and peak there: $\Omega(N, N/2) = 1.85 \times 10^5,\ 1.26 \times 10^{14},\ 1.08 \times 10^{23},\ 1.01 \times 10^{29}$ and $P_{N/2} = 0.176,\ 0.112,\ 0.089,\ 0.080$. For $N = 100$, the total is $2^{100} = 1.27 \times 10^{30}$. Because Ω differs by many orders of magnitude between sizes, on a linear axis only the largest curve would be visible; hence the log scale and the normalised probability.

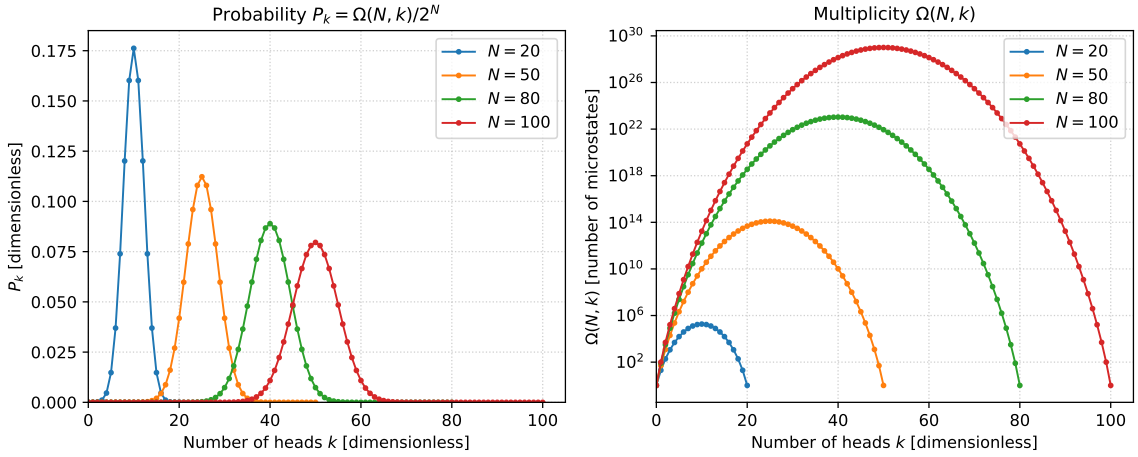


Figure 2: Left: probability P_k versus number of heads k (dimensionless). Right: multiplicity $\Omega(N,k)$ (number of microstates, log scale). Values are exact to $\sim 10^{-13}$ relative.

3.4 Largest N for the distribution

Computing the $N+1$ coefficients with the recurrence $\Omega(N,0) = 1$, $\Omega(N,k+1) = \Omega(N,k)(N-k)/(k+1)$ in exact integers avoids generating 2^N states. Two tests (300 s limit each) gave different limits. *Calculation:* all coefficients were obtained up to $N = 354\,419$, beyond which the RAM was exhausted, since each coefficient is an integer of $\sim N$ bits and the total memory grows $\propto N^2$. *Plotting:* the largest size plotted without error was $N = 1028$, because converting the integers to double precision overflows above $\sim 1.8 \times 10^{308}$ (the exact threshold for $\Omega(N, N/2)$ is $N = 1029$). This limit is not fundamental: computing in log-space (2) avoids the overflow entirely, and the accuracy then limits N only through rounding ($|\sum_k P_k - 1| = 6 \times 10^{-13}$ at $N = 10^3$ and 1×10^{-8} at $N = 10^7$, with ≈ 80 MB per array at $N = 10^7$). The practical maximum is therefore $N \sim 10^7$ in double precision, limited by memory and accuracy, versus $N = 1028$ for direct integer-to-float conversion.

3.5 Thermodynamic limit

For N fair coins k is binomially distributed with $\mu = N/2$ and $\sigma = \sqrt{N}/2$. By Stirling's formula, with $x = k - N/2$ and $|x| \ll N$,

$$P_k \simeq \sqrt{\frac{2}{\pi N}}\, e^{-2x^2/N}, \qquad \frac{\sigma}{\mu} = \frac{1}{\sqrt{N}}, \qquad \ln \Omega\left(N, \tfrac{N}{2} + x\right) \simeq N \ln 2 + \tfrac{1}{2} \ln \frac{2}{\pi N} - \frac{2x^2}{N}. \tag{3}$$

For $N = 10^{23}$: $\mu = 5 \times 10^{22}$, $\sigma = 1.58 \times 10^{11}$ heads, and $\sigma/\mu = 3.16 \times 10^{-12}$. The absolute fluctuation is huge but the relative one is negligible: the distribution is a spike at $k/N = \frac{1}{2}$ (a deviation $k/N = \frac{1}{2} + \delta$ has probability $\sim e^{-2N\delta^2}$), so the macroscopic state is sharply defined although each coin is random. Since $S = k_\mathrm{B} \ln \Omega$ is largest at $k = N/2$, the equilibrium macrostate coincides with the maximum-entropy one, and $\Delta S \propto N k_\mathrm{B}$ between it and ordered states is macroscopic.

4 Part 2: probabilities, entropy and thermodynamics

The team size is $N = 100 + M$, where M is the harmonic mean of the last two digits of the team members' NIAs (92, 75, 99 and 14):

$$M = \frac{n}{\sum_{i=1}^{n} \frac{1}{x_i}} = \frac{4}{\frac{1}{92} + \frac{1}{75} + \frac{1}{99} + \frac{1}{14}} = 37.83 \simeq 38, \tag{4}$$

with $n = 4$ members and x_i the last two NIA digits. Rounding to the nearest integer gives $M = 38$ and $N = 138 < 200$. No digit is zero, so no replacement was needed in the harmonic mean. As N is even, $N/2 = 69$ is an integer. All quantities are computed in log-space with $\ln \Omega(N,k) = \ln \Gamma(N+1) - \ln \Gamma(k+1) - \ln \Gamma(N-k+1)$, which avoids overflow ($2^{138} = 3.48 \times 10^{41}$, $\Omega(138, 69) = 2.36 \times 10^{40}$). They are exact up to floating-point rounding, which is the only source of uncertainty and is quantified by comparison with exact integer arithmetic (`math.comb`).

4.1 Probabilities

Table 2 gives $P_k = \Omega(N,k)/2^N$ for items a) and b), and the cumulative probability $\Delta P_{k_1 k_2} = \sum_{k=k_1}^{k_2} P_k$ for items c) and d). Since the distribution is symmetric under $k \to N - k$, a range of $N/2 \pm 20$ *tails* (item d) is the same as $N/2 \pm 20$ heads.

item	quantity	range	value
a)	P_{30}	$k = 30$	5.650×10^{-12}
b)	P_{69} (most probable)	$k = N/2$	0.0678
c)	ΔP	$59 \le k \le 79$	0.926549
d)	ΔP	$49 \le k \le 89$	0.999559

Table 2: Probabilities for $N = 138$ (dimensionless). Relative uncertainty (rounding) $\lesssim 10^{-13}$: the largest deviation from exact integer arithmetic is 9.6×10^{-14} in the four items and 2.1×10^{-13} over all k; $\sum_{k=0}^{N} P_k = 1 - 4.3 \times 10^{-14}$.

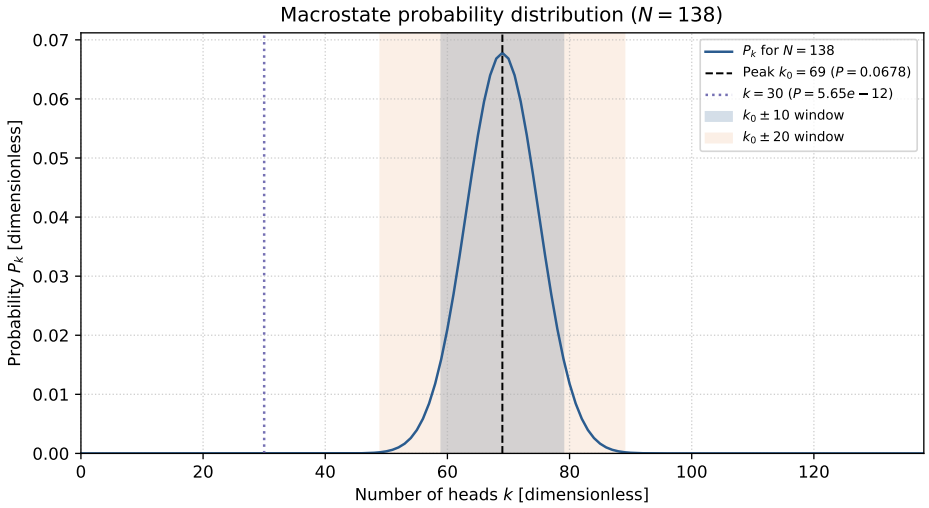


Figure 3: Macrostate probability P_k for $N = 138$ coins (dimensionless). Dashed: most probable macrostate $k_0 = 69$; shaded: $k_0 \pm 10$ (dark) and $k_0 \pm 20$ (light) windows; dotted: $k = 30$.

The macrostate with 30 heads is $P_{69}/P_{30} = 1.2 \times 10^{10}$ times less probable than the peak, only because it has fewer microstates ($\Omega = 1.97 \times 10^{30}$ versus 2.36×10^{40}); each microstate has the same probability 2^{-138}. The peak holds only 6.8% of the probability, but the ± 10 window (1.7σ,

$\sigma = 5.87$) holds 92.7% and the ± 20 window (3.4σ) 99.96%, in agreement with the Gaussian (3) (0.9262, 0.99952).

4.2 Entropy changes

We compute $\Delta S_{k_1 k_2} \equiv S(k_1) - S(k_2)$, so that $\Delta S/k_\mathrm{B} = \ln[\Omega(N, k_1)/\Omega(N, k_2)]$, with $k_\mathrm{B} = 1.380649 \times 10^{-23}$ J/K (exact SI value); results are in Table 3.

item	(k_1, k_2)	states	$\Delta S/k_\mathrm{B}$	ΔS (J/K)	Ω_1/Ω_2
e)	$(94, 69)$	high disorder $\to$ maximum	-9.1974	-1.270×10^{-22}	1.013×10^{-4}
f)	$(79, 69)$	moderate $\to$ maximum	-1.4438	-1.993×10^{-23}	0.2360
g)	$(69, 20)$	maximum $\to$ high order	$+38.2003$	$+5.274 \times 10^{-22}$	3.892×10^{16}

Table 3: Entropy differences for $N = 138$ (numerical uncertainty $< 10^{-13}$ in $\Delta S/k_\mathrm{B}$).

Table 3 shows that the maximum entropy is at $k = N/2$: moving away from it reduces S. In cases e) and f), $S(k_1) < S(k_2)$ and hence $\Delta S < 0$. In case g), $S(k_1) > S(k_2)$ and $\Delta S > 0$, with a much larger magnitude because $k = 20$ is far from the peak ($\Omega_1/\Omega_2 \sim 10^{16}$).

4.3 Thermodynamic implications

h) Time to a highly ordered state. We assume a tossing rate of N coins per second, i.e. one independent trial of the whole system per second, and the probability of success is $P_8 = \Omega(138, 8)/2^{138} = 7.61 \times 10^{-30}$. The number of trials until the first success is geometric with mean

$$\langle t \rangle = \frac{1}{P_8} = 1.31 \times 10^{29} \text{ s} = 4.2 \times 10^{21} \text{ yr} \approx 3 \times 10^{11} \text{ ages of the universe} \tag{5}$$

i) Likelihood ratio. The ratio between the most probable macrostate and another one is

$$\frac{P_{k_a}}{P_{k_b}} = \frac{\Omega(N, k_a)}{\Omega(N, k_b)} = e^{\Delta S_{k_a k_b}/k_\mathrm{B}}. \tag{6}$$

For $M \pm 10$ heads ($k = 48$ and $k = 28$, with $M = 38$) the ratios are 629.8 and 1.65×10^{11}, respectively.

j) Second Law. None of the obtained results contradicts the Second Law. Statistically, an isolated system evolves towards macrostates of larger multiplicity because they are more probable, not because decreases of S are forbidden. The transitions obtained for the analysed entropy changes are improbable but possible. An example is case g), where the likelihood ratio of the move from the maximum to $k = 20$ is $e^{-38.2} = 2.6 \times 10^{-17}$, which shows that it is very unlikely to be observed. In contrast, for case f) $\Delta S/k_\mathrm{B} = -1.4$, which corresponds to visible fluctuations. This way, it can be reaffirmed that the Second Law of Thermodynamics holds for these cases.

4.4 Large-N limit ($N \to 10^{23}$)

All items are re-evaluated for $N = 10^{23}$ with the Gaussian form (3), and with $\ln \binom{N}{k} = \sum_{j=0}^{k-1} \ln[(N-j)/(k-j)]$ for small k (items a and h). The results are in Table 4.

item	result for $N = 10^{23}$ ($\sigma = 1.58 \times 10^{11}$)
a) P_{30}	$\log_{10} P_{30} = -3.01 \times 10^{22}$ (zero in practice)
b) $P_{N/2}$	2.52×10^{-12}
c), d) ΔP, $N/2 \pm 10$, ± 20	5.30×10^{-11}, 1.03×10^{-10}
e), f) $\Delta S/k_\mathrm{B}$	-1.25×10^{-20}, -2.0×10^{-21}
g) ΔS	$6.93 \times 10^{22}\, k_\mathrm{B} \simeq N k_\mathrm{B} \ln 2 = 0.957$ J/K
h) waiting time	$\log_{10} t = 3.01 \times 10^{22}$ (t in s)
i) likelihood ratio	$P_{N/2}/P_{N/2\pm 10} - 1 = 2.0 \times 10^{-21}$

Table 4: Items a–i for $N = 10^{23}$ (analytic; relative error of $\ln \Omega$ is $O(1/N)$).

Entropy differences between nearby macrostates vanish, whereas the difference to an ordered state is extensive, $\simeq N k_\mathrm{B} \ln 2 \approx 0.96$ J/K, and the waiting time is $10^{3\times 10^{22}}$ s. As a result, the Second Law becomes deterministic for $N \to \infty$, since the relative fluctuations scale as $N^{-1/2} \to 0$.

5 Conclusions

The results obtained for the system of coins agree with the statistical picture described in the manual. In the enumeration of microstates, storing every state is limited by memory ($N = 27$ was the last size completed, and $N = 28$ exhausted the RAM), whereas the generator only keeps one counter per macrostate, so its cost is dominated by time, which grows by 2^N and gives the extrapolated limit $N_\mathrm{max} \approx 31$ for five minutes. For the multiplicities, exact integers reached $N = 354,419$ in the calculation but only $N = 1028$ in the plot, because of the double-precision overflow; computing in log-space removes this limit, and the practical maximum, set by memory and rounding, is $N \sim 10^7$.

For the team size $N = 138$, the distribution is symmetric and peaked at $k = N/2 = 69$ ($P_{69} = 0.0678$), and the probability is concentrated around it: 92.7% lies within ± 10 heads and 99.96% within ± 20, in agreement with the Gaussian approximation. States far from the peak, such as $k = 30$, are less probable only because they have fewer microstates, since every microstate has the same probability 2^{-138}. The entropy is maximal at $k = N/2$, and the ordered macrostate with 20 heads has $38.2\, k_\mathrm{B}$ less entropy. Reaching a state of 8 heads by random tossing would take $\sim 4 \times 10^{21}$ years, so such low-entropy states are not forbidden but extremely improbable, and the Second Law is not contradicted by any of the results. For $N \sim 10^{23}$ the relative width of the distribution is $N^{-1/2} = 3.2 \times 10^{-12}$ and the entropy gap to an ordered state is extensive, $\simeq N k_\mathrm{B} \ln 2 \approx 0.96$ J/K, so the macroscopic state is sharply defined and the Second Law becomes deterministic.

A All microstates for $N = 7$

All $2^7 = 128$ microstates are listed in Table 5, grouped by number of heads k, as in Table 1 of the lab manual.

heads k	tails $7-k$	$\Omega(7,k)$	microstates
7	0	1	HHHHHHH
6	1	7	THHHHHH HTHHHHH HHTHHHH HHHTHHH HHHHTHH HHHHHTH HHHHHHT
5	2	21	TTHHHHH THTHHHH THHTHHH THHHTHH THHHHTH THHHHHT HTTHHHH HTHTHHH HTHHTHH HTHHHTH HTHHHHT HHTTHHH HHTHTHH HHTHHTH HHHTTHH HHHTHTH HHHTHHT HHHHTTH HHHHTHT HHHHHTT
4	3	35	TTTHHHH TTHTHHH TTHHTHH TTHHHTH TTHHHHT THTTHHH THTHTHH THTHHTH THTHHHT THHTTHH THHTHTH THHTHHT THHHTTH HTTTHHH HTTHTHH HTTHHTH HTTHHHT HTHTTHH HTHTHTH HTHTHHT HTHHTTH HTHHTHT HTHHHTT HHTTTHH HHTTHTH HHTTHHT HHTHTTH HHTHTHT HHTHHTT HHHTTTH HHHTTHT HHHTHTT HHHHTTT
3	4	35	TTTTHHH TTTHTHH TTTHHTH TTTHHHT TTHTTHH TTHTHTH TTHTHHT TTHHTTH TTHHTHT TTHHHTT THTTTHH THTTHTH THTTHHT THTHTTH THTHTHT THTHHTT THHTTTH THHTTHT THHTHTT THHHTTT HTTTTHH HTTTHTH HTTTHHT HTTHTTH HTTHTHT HTTHHTT HTHTTTH HTHTTHT HTHTHTT HTHHTTT HHTTTTH HHTTTHT HHTTHTT HHTHTTT HHHTTTT
2	5	21	TTTTTHH TTTTHTH TTTTHHT TTTHTTH TTTHTHT TTTHHTT TTHTTTH TTHTTHT TTHTHTT TTHHTTT THTTTTH THTTTHT THTTHTT THTHTTT THHTTTT HTTTTTH HTTTTHT HTTTHTT HTTHTTT HTHTTTT HHTTTTT
1	6	7	TTTTTTH TTTTTHT TTTTHTT TTTHTTT TTHTTTT THTTTTT HTTTTTT
0	7	1	TTTTTTT

Table 5: All $2^7 = 128$ microstates of $N = 7$ coins, one row per macrostate (H = heads, T = tails); $\Omega(7,k)$ is the number of microstates in each row and the rows add up to 128.

B AI collaboration statement

AI assistants (Claude, Gemini and Copilot) were used for drafting code, code redefinition and proper implementation of mathematical procedures and contraints. Additionally, the AI assistant were applied for the elaboration of some comments and to look for Git Hub implementation commands. Every prompt, generated code and modification is logged in `AI_STATEMENT.md`. All AI-generated logic is covered by tests of each section and was it reviewed by the corresponding team member who implemented it.

C Github link

In this section, the link to the Github was inserted: https://github.com/100542175/LAB-1-STATISTICAL-PHYSICS